DS-265 DLCV 2026: Assignment 2

Vision Transformer Analysis on CIFAR-10

Rohit Singh (26119)

1 Experimental Setup

ViT: Patch size 4, embedding dim 256, 6 Transformer blocks, 8 heads, MLP hidden dim 1024, learnable CLS token (appended), pre-norm with GELU, dropout 0.1. ∼**4.77 M** params.

ResNet: Custom 3-stage design. 5×5 projection conv $\rightarrow$ 3 stages of 10 residual blocks each (30 total), 5×5 kernels, BatchNorm+ReLU. Channels double per stage ($15 \rightarrow 30 \rightarrow 60 \rightarrow 120$) via stride-2 convs. Final BN $\rightarrow$ FFN(512) $\rightarrow$ Linear(10). ∼**4.67 M** params (≈0.1M less than ViT).

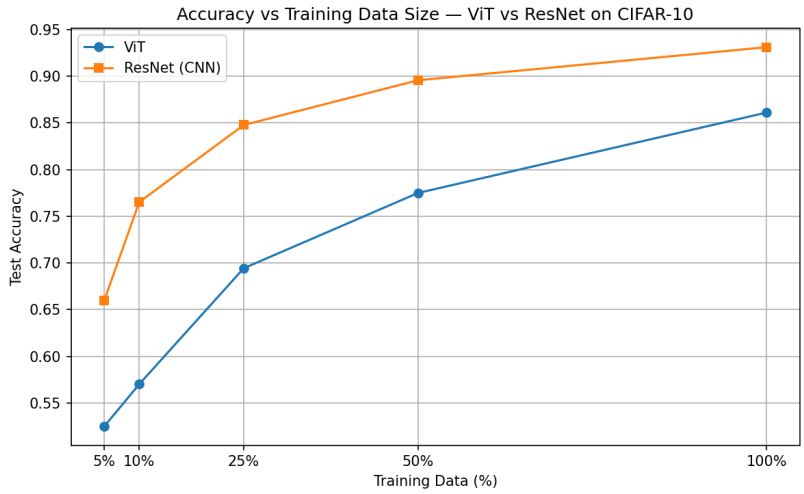
Training (common): AdamW (lr=10^{-3}, wd=0.1), 10-epoch linear warmup + cosine decay, 200 epochs, batch 512, bfloat16 mixed precision, grad clip 1.0. Augmentation: random crop (pad=4), horizontal flip, ColorJitter(0.1), MixUp(α=0.2), label smoothing 0.1, CIFAR-10 normalization.

Both models have roughly equal parameter counts so accuracy comparisons are fair w.r.t. capacity.

2 Experiment 1: Data Efficiency — ViT vs CNN

We train both on $\{5, 10, 25, 50, 100\}\%$ of CIFAR-10 training data.

Frac.	ViT	ResNet	Gap
5%	0.5251	0.6601	13.5%
10%	0.5701	0.7651	19.5%
25%	0.6943	0.8477	15.3%
50%	0.7749	0.8956	12.1%
100%	0.8610	0.9310	7.0%



ResNet outperforms ViT across all splits. Since both have nearly equal parameters, the gap comes from inductive biases — CNNs get translation equivariance and locality for free via weight-shared convolutions, while ViT has to learn spatial structure entirely from data. Looking at the loss curves, ViT overfits noticeably at lower fractions — training loss keeps dropping while test accuracy plateaus early, especially at 5% and 10%. ResNet shows much less of this train-test divergence, consistent with weight-sharing acting as implicit regularization.

The gap narrows from 13.5% to 7.0% as data grows, suggesting ViT could catch up with more data or pretraining, consistent with the original ViT paper's findings on JFT-300M. Interestingly, the gap peaks at 10% (19.5%) rather than 5% — at 5% (2500 samples) both models struggle enough that the gap is smaller, but at 10% ResNet's inductive biases start paying off strongly while ViT still can't leverage the data well. Beyond 10% the gap steadily closes as the ViT gets enough examples to learn spatial structure on its own.

3 Experiment 2: Effect of Patch Size

Patch sizes $\{4, 8, 16\}$, 3 runs each. All other hyperparams fixed.

Patch	Avg Acc	Std	Time (s)
4	0.8574	0.0029	1102.9
8	0.8187	0.0069	472.8
16	0.7268	0.0025	464.4

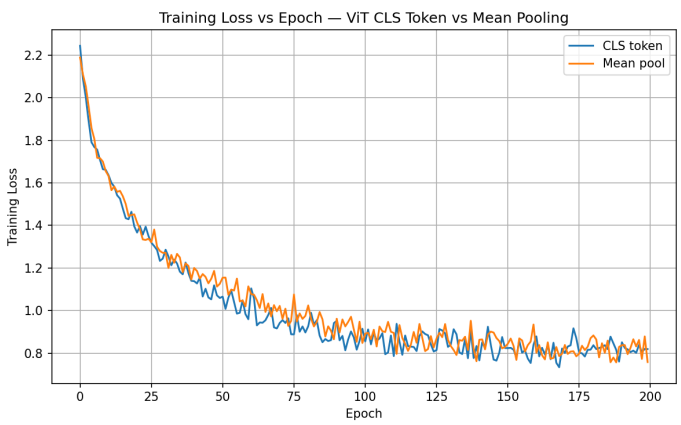
Accuracy drops sharply with larger patches. On 32×32 images, patch 16 gives only $(32/16)^2 = 4$ tokens vs 16 for patch 8 and 64 for patch 4. Larger patches compress more spatial detail into a single embedding, losing fine-grained info at patchification. With only 4 tokens, attention can't do much meaningful spatial reasoning either — it's basically a 4-element sequence, which limits the model's ability to capture inter-region dependencies.

An interesting detail: patch 8 has the highest run-to-run variance (std 0.0069), roughly 2× that of patch 4 and patch 16. We think this is because 16 tokens is a "middle ground" where the model is sensitive to initialization — enough tokens to learn non-trivial attention patterns, but not enough for those patterns to be robust across random seeds. Patch 4 has enough tokens (64) for stable learning, and patch 16 has so few (4) that the model converges to a similar simple solution regardless of init.

Patch 4 takes ∼2.3× longer than patch 8, reflecting quadratic attention cost (64^2 vs 16^2). Patch 8 and 16 have nearly identical times because at 16 and 4 tokens, attention is trivially cheap and the bottleneck shifts to data loading and the per-token MLP.

4 Experiment 3: CLS Token vs Mean Pooling

Pooling	Best Acc	Time (s)
CLS token	0.8638	1555.6
Mean pool	0.8653	1349.0



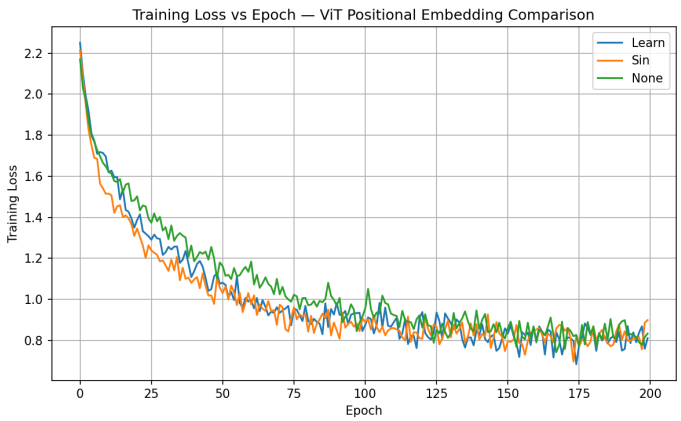
Essentially identical results. With mean pooling the model spreads class info across all patch tokens instead of funneling it into a single CLS token, and gradient flow during backprop is more uniform

across the sequence. The loss curves show both converge at similar rates, with mean pooling being marginally smoother in early epochs — likely because averaging over 64 tokens gives a more stable gradient signal than relying on a single token.

The ViT paper noted mean pooling can sometimes need a different lr, but we got equivalent results at the same lr — likely our warmup+cosine schedule is adaptive enough. This also suggests that at CIFAR-10 scale (small images, 10 classes), the aggregation strategy just doesn't matter much — there's not enough complexity for one to meaningfully outperform the other.

5 Experiment 4: Positional Encoding Ablation

Pos. Embed	Best Acc	Time (s)
Learnable	0.8543	1946.8
Sinusoidal	0.8718	1360.1
None	0.8071	1188.0



Sinusoidal slightly beats learnable, which is a bit surprising since learnable has strictly more capacity. But with only 64 positions, learnable embeddings add $64 \times 256 = 16$k extra free parameters that can overfit. Sinusoidal gives reasonable positional structure without any learnable params — acts as implicit regularization. We didn't run multiple seeds so the gap may not be significant; both used identical hyperparams.

As the ViT paper notes, since our Transformer operates on patch-level inputs (not pixels), the positional encoding scheme matters less — the grid is only 8×8 and each patch already spans a 4×4 spatial region.

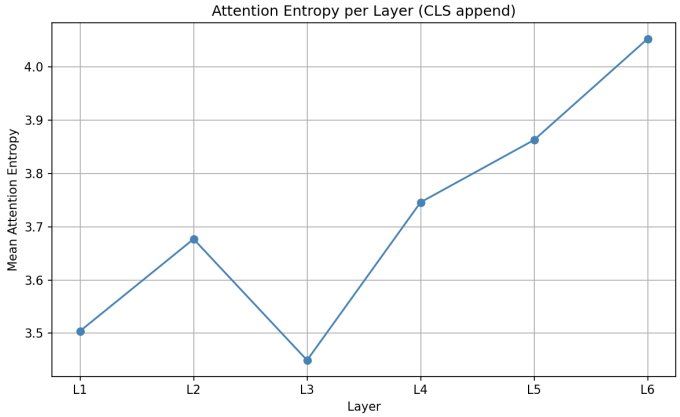
No positional encoding still hits 80.7%. CIFAR-10 classification cares more about *what* features are present than *where* they are — patch embeddings already encode content, and attention can route by content similarity alone. Still, the 5–7% gap shows position does provide useful signal. Looking at the loss curves, the no-position variant converges faster initially (fewer params to learn) but plateaus earlier and at a lower accuracy, suggesting it hits a representational ceiling where spatial structure would help.

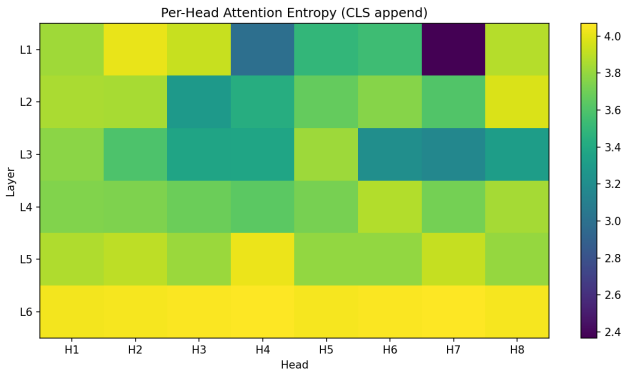
6 Experiment 5: Attention Entropy Analysis

We compute $H = -\sum_j p_j \log p_j$ per head. Max entropy for 65 tokens (64 patches + CLS): $\ln(65) \approx 4.17$.

Layer	Mean H	Layer	Mean H
1	3.504	4	3.746
2	3.677	5	3.863
3	3.450	6	4.053

Lyr	H1	H2	H3	H4	H5	H6	H7	H8
1	3.82	4.01	3.93	2.99	3.49	3.54	2.37	3.88
3	3.77	3.60	3.36	3.37	3.82	3.21	3.16	3.32
6	4.04	4.05	4.06	4.06	4.04	4.06	4.07	4.05





Entropy broadly increases with depth (3.50 → 4.05) — deeper layers attend more uniformly, doing global aggregation. Layer 6 heads are all ∼4.05 (within 3% of max entropy), so every token attends to nearly everything equally.

Layer 3 is the most selective — lowest mean entropy (3.45), highest head variance (range 3.16–3.82). This is where content-specific attention happens; some heads focus on particular spatial regions while others stay broad. This makes it the "workhorse" layer of the network — it's doing the most structured feature extraction rather than just mixing everything together.

Layer 1 also shows high head diversity (range 2.37–4.01). Head 7 stands out with $H = 2.37$, meaning it concentrates attention on just a few tokens — likely acting as a local or edge-detecting feature extractor. Meanwhile Head 2 in the same layer is near-uniform ($H = 4.01$), already doing global aggregation. This kind of specialization within a layer is useful: different heads can capture patterns at different spatial scales simultaneously.

Head specialization collapses with depth: the entropy spread goes from 1.64 in L1 to just 0.03 in L6. By the final layer, all 8 heads converge to nearly identical uniform attention. This suggests the model has already extracted all the discriminative features it needs by the middle layers, and the last layers just aggregate everything into a consensus representation for classification. It's somewhat wasteful — the 8 heads in L6 are basically redundant. This hints that for CIFAR-10, fewer heads or layers might suffice in the final block.

Attention map visualizations. Figure 1 shows CLS attention (mean over heads) reshaped to the 8×8 patch grid, overlaid on test images across all 6 layers. The maps confirm the entropy analysis: L1–L2 attention is diffuse, L3–L4 sharpens onto discriminative object regions (dog's face, truck cabin, airplane fuselage against sky), and L5–L6 spreads out again. This matches the entropy dip at Layer 3 — selective heads zero in on task-relevant parts, then later layers aggregate globally for classification.

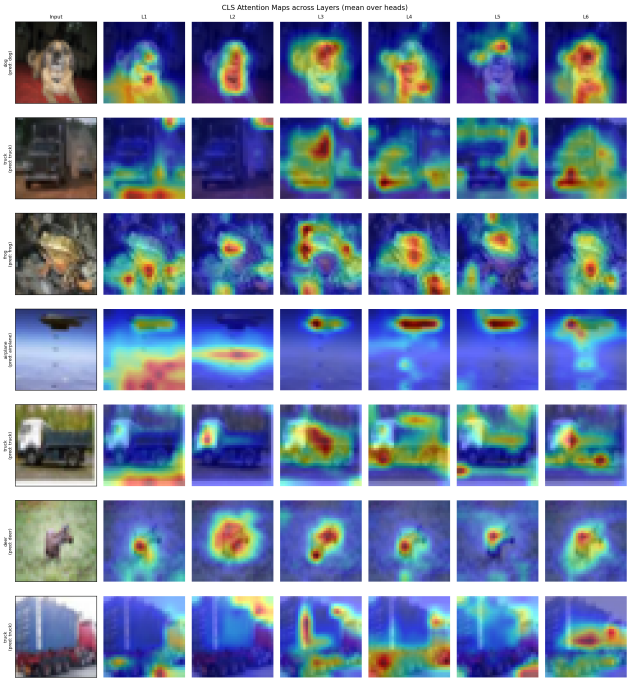


Figure 1: CLS attention maps across layers (mean over heads).

7 Experiment 6: Overlapping vs Non-Overlapping Patches

Stride = patch_size (non-overlap, 64 tokens) vs stride = patch_size/2 (overlap). With patch 4, stride 2: $(32-4)/2+1=15$, so 225 tokens ($\approx$3.5$\times$ more).

	Acc	**Time (s)**	**Mem (MB)**
Non-overlap	0.8594	1278.5	2663
Overlap	0.8815	3787.9	12359

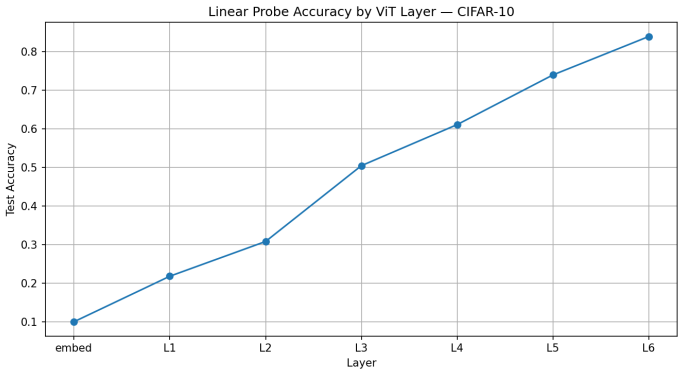
Overlap gives +2.2% accuracy. Adjacent patches share 75% of their area (stride 2 on a 4×4 patch), providing a denser, more redundant view of the image that captures spatial transitions at patch boundaries which non-overlapping patches miss entirely. The model sees every local region from multiple overlapping perspectives, which is conceptually similar to how CNNs with stride $<$ kernel size work.

The cost is steep: 3$\times$ slower and 4.6$\times$ more memory. The memory ratio exceeds the 3.5$\times$ token ratio because attention scales quadratically — the attention matrix goes from 64^2 to 225^2 ($\approx$12.4$\times$), which dominates the memory profile even though MLP and embeddings scale linearly. For CIFAR-10's small images the tradeoff is acceptable, but on 224×224 images this approach would be impractical without efficient attention.

8 Experiment 7: Layer-wise Representation Quality

We freeze the trained ViT, extract the CLS token from each layer, and train a linear classifier on it for 100 epochs.

Layer	Acc	Layer	Acc
Embed	0.1007	L4	0.6109
L1	0.2185	L5	0.7395
L2	0.3084	L6	0.8389
L3	0.5046		



Accuracy goes from chance (10.1%) to 83.9% monotonically. Embedding layer has zero class info as expected — it's just a linear projection of pixel patches with no inter-patch interaction yet.

The biggest jump is L2→L3 (+19.7%). This connects directly to Experiment 5: Layer 3 had the most selective attention (lowest entropy, highest head diversity). The selective heads in L3 are apparently focusing on discriminative image regions, producing representations that cross the linear separability threshold for many classes. This is the clearest evidence that *attention selectivity drives feature quality* — the layer doing the most structured, non-uniform attention is also the layer contributing the most to linear probe accuracy.

The progression is roughly linear (∼12–15% per layer after L1), meaning every layer contributes meaningfully. There's no single "magic" layer; the Transformer builds up class-discriminative features gradually through repeated attention and MLP processing.

L6 probe gets 83.9% vs the full model's 86.1%. The ∼2% gap comes from the probe not using LayerNorm before classification (the full model does) and training for 100 vs 200 epochs with a simpler schedule (no warmup, no MixUp). Still, 83.9% confirms the CLS representation is already high-quality by the final layer — the classification head on top adds relatively little.

9 Conclusion

CNNs are more data-efficient than ViTs without pretraining, but the gap narrows with more data. Smaller patches give better accuracy at quadratic cost. CLS and mean pooling are equivalent at CIFAR-10 scale. Positional encodings help (∼5–7%), with sinusoidal slightly ahead, possibly due to regularization from having no learnable positional params. Entropy analysis reveals a clear transition from selective, specialized heads in early/mid layers to uniform aggregation deeper, with Layer 3 being the most selective — and also where linear probe accuracy jumps the most, tying together Experiments 5 and 7. Overlapping patches trade substantial compute for a modest accuracy gain by providing a denser spatial representation to attention.